

Sina Yari

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| 🌐 <https://sinayari.github.io/> | **Permanent Resident of the USA (Green Card Holder)**

I am a Ph.D. student and Graduate Research Assistant in Computer Science at George Mason University, where I develop optimization- and security-aware scheduling techniques to mitigate schedule-based side-channel attacks in safety-critical real-time embedded systems. Through this research, I am gaining hands-on experience in algorithm design, mathematical modeling, large-scale simulation, machine learning, artificial intelligence, and optimization. I implement and evaluate these techniques in Python and C/C++, bridging advanced research ideas with practical engineering solutions.

TECHNICAL SKILLS

Programming Languages & Frameworks: C/C++, Java, Python, SQL, Bash, NumPy, Pandas, Embedded C, MATLAB, Verilog, VHDL
DevOps Tools: Git
Data Base Systems: MySQL (Relational Database)
AI & Machine Learning: Reinforcement Learning
Simulation: Quartus, Xilinx, Gem5, McPAT, HotSpot, Design Compiler, Hspice, Pspice
Microcontrollers and FPGA Boards: Raspberry Pi, ARM Cortex, AVR, STM, NXP LPC, DE1, DE2, Spartan

EDUCATION

George Mason University <i>PhD in Computer Science (GPA: 3.8 / 4.0)</i>	Fairfax, VA Aug 2022 – Present (Expected: Aug 2027)
Sharif University of Technology <i>Master of Science in Computer Engineering - Computer Systems Architecture (GPA: 4.0 / 4.0)</i>	Tehran, Iran 2018 – 2021
Ferdowsi University of Mashhad <i>Bachelor of Science in Computer Engineering - Hardware</i>	Mashhad, Iran 2012 – 2017

PROFESSIONAL EXPERIENCE

Graduate Research Assistant <i>George Mason University (Systems and Networking Group)</i>	Aug 2022 – Present Fairfax, VA
- Designed and implemented security-aware scheduling techniques for real-time embedded systems to mitigate schedule-based side-channel attacks; published in ISORC 2026 and JSA 2023. - Developed large-scale C++ and Python simulation frameworks to evaluate real-time scheduling algorithms.	
Graduate Teaching Assistant <i>George Mason University</i>	Aug 2022 – Present Fairfax, VA
- Program Design and Data Structures (Java) - Database Systems - Intro Programming (Python) - Computer Systems and Programming (C & Assembly)	- Computer Communications and Networking - SWE for Real-Time Embedded Systems - Advanced Computer Architecture - Computer Systems Architecture
Graduate Research Assistant <i>Sharif University of Technology (Embedded Systems Research Laboratory)</i>	2018 – 2021 Tehran, Iran
- Designed energy-, thermal-, reliability-, and aging-aware scheduling techniques for real-time embedded systems to improve efficiency, safety, and fault tolerance. - Developed and analyzed system-level optimization techniques, and explored reinforcement learning-based approaches for real-time embedded systems.	
Graduate Teaching Assistant <i>Sharif University of Technology</i>	2018 – 2021 Tehran, Iran
- Fault-Tolerant Systems Design - Low Power Design - Digital Systems Design - Numerical Methods	- Embedded Systems - Real-Time Systems - Logical Circuits Laboratory (Instructor) - Digital Systems Design Laboratory (Instructor)
Undergraduate Research Assistant <i>Ferdowsi University of Mashhad Dependable Distributed Embedded Systems Laborator</i>	2014 – 2017 Mashhad, Iran
Analyzed the $\mu\text{C}/\text{OS-III}$ task scheduler in C for an ARM Cortex-M3-based real-time embedded system.	
Undergraduate Teaching Assistant <i>Ferdowsi University of Mashhad</i>	2014 – 2017 Mashhad, Iran
- Logical Circuits - Machine and Assembly Languages - Computer Architecture	

HONORS & AWARDS

Student Travel Grant, ACM SIGMETRICS 2025

Stony Brook, NY, USA

- Awarded for accepted research poster on real-time systems security

Certificate in Inclusive Teaching for STEM Graduate Students

George Mason University, VA, USA

- Completed certified training on inclusive and effective STEM teaching practices

Graduate Teaching Assistant (GTA)

George Mason University, VA, USA

- Led recitation sessions and office hours to clarify course concepts and support student success
- Assisted with grading assignments and projects for undergraduate and graduate-level courses

Top 5 M.Sc. Student, Computer Systems Architecture

Sharif University of Technology, Tehran, Iran

- Ranked 5th among 27 students

Top 0.1%, Nationwide M.Sc. Entrance Exam

Iran

- Ranked 30th out of ~30,000 participants in the Computer Engineering entrance exam (2018)

PUBLICATIONS

S. Yari Karin, H. Aydin, and D. Zhu, "Task Splitting to Mitigate Schedule-Based Anterior Attacks in Real-Time Embedded Systems," *29th International Symposium on Real-Time Distributed Systems (ISORC)*, 2026.

S. Yari Karin, H. Aydin, D. Zhu, S. Drager, and M. Anderson, "Impact of Priority Assignment on Schedule-Based Attacks in Real-Time Embedded Systems," *Journal of Systems Architecture*, 2023.

S. Yari Karin, R. Siyadatzaheh, M. Ansari, and A. Ejlali, "Passive Primary/Backup-Based Scheduling for Simultaneous Power and Reliability Management on Heterogeneous Embedded Systems," *IEEE Transactions on Sustainable Computing*, 2022.

S. Safari, M. Ansari, H. Khdr, P. Gohari-Nazari, S. Yari Karin, A. Yeganeh-Khaksar, S. Hessabi, A. Ejlali, and J. Henkel, "A Survey of Fault-Tolerance Techniques for Embedded Systems from the Perspective of Power, Energy, and Thermal Issues," *IEEE Access*, 2022.

M. Ansari, S. Safari, S. Yari Karin, P. Gohari-Nazari, H. Khdr, M. Shafique, J. Henkel, and A. Ejlali, "Thermal-Aware Standby-Sparing Technique on Heterogeneous Real-Time Embedded Systems," *IEEE Transactions on Emerging Topics in Computing*, 2021.

M. Ansari, S. Yari Karin, S. Safari, and A. Ejlali, "Power Management to Meet Thermal Safe Power in Fault-Tolerant Embedded Systems," *TechRxiv*, 2021.

A. Yeganeh-Khaksar, M. Ansari, S. Safari, S. Yari Karin, and A. Ejlali, "Ring-DVFS: Reliability-Aware Reinforcement Learning-Based DVFS for Real-Time Embedded Systems," *IEEE Embedded Systems Letters*, 2021.

J. Saber-Latibari, M. Ansari, P. Gohari-Nazari, S. Yari Karin, A. M. H. Monazzah, and A. Ejlali, "READY: Reliability- and Deadline-Aware Power-Budgeting for Heterogeneous Multicore Systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2021.

S. Yari Karin, A. Sahraee, J. Saber-Latibari, M. Ansari, N. Rohbani, and A. Ejlali, "A Comparative Study of Joint Power and Reliability Management Techniques in Multicore Embedded Systems," *CSI/CPSSI International Symposium on Real-Time and Embedded Systems and Technologies (RTEST)*, 2020.